SFE2D ver. 2.0

for windows 7/8/10

Full User Manual

SFE2D: A 2D spatial & spectral feature extraction tool for geo-imaging

Using the unsupervised source separation methods like

principal component analysis (variance maximization), independent component analysis (kurtosis and negentropy maximization), continuous wavelet transform, RGB color processing, and k-mean clustering segmentation for feature extraction and dimensionality reduction of hyperdimensional geoscientific data sets.

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1. Introduction

SFE2D is a standalone 2D spectral feature extraction program based on integration of principal component analysis (PCA), independent component analysis (ICA), and continuous wavelet transform (CWT). The program offers spatial and spectral source separation for multiple sources of geoscientific data sets. SFE2D also provides a graphic user interface with the ability to demonstrate compiled features in the form of RGB images as well as segmented images through a fast k-mean clustering algorithm to reduce the number of colors and compile pseudo-geologic maps from extracted features. Users can also pick a specified color range based on manual selection of the specified color intensities with mouse clicks. The program's application is straightforward and for immediate use, as long as users already installed Matlab runtime library (free to download and install) on their computer.

2. System requirements

This program is designed to run on any windows based personal computer with at least 8 GB of random-access memory (RAM). Increasing the RAM size allows larger images to be processed at once. Since large matrices are operating in this program, the read/write speed of the storage is also essential. A solid-state drive (SSD) with a non-volatile memory express interface (NVMe) is recommended.

The SFE2D is provided in two formats, including Matlab P-codes and a standalone executable program.

- P-codes require Matlab 2021a version 9.10 (and later) to run. To do so, one needs to locate the P-codes in the current folder of Matlab and then types SFE2D in Matlab Command Window.

- To use the executable program (SFE2D.exe) user does not need any previously installed Matlab software on the PC. The only prerequisite is installing

the latest Matlab 2021a Runtime library that is free to download from *mathworks.com* (version 9.10). To do so, users can install the required runtime codes by double-clicking on the offline program installer "Installer_SFE2D (Offline).exe" or the online version of it "Installer_SFE2D (Online).exe" that is a lightweight version in which the program and the runtime codes are going to be installed automatically through downloading from MathWorks server.

Either way, after the double click, the following windows pop up:

Figure 1

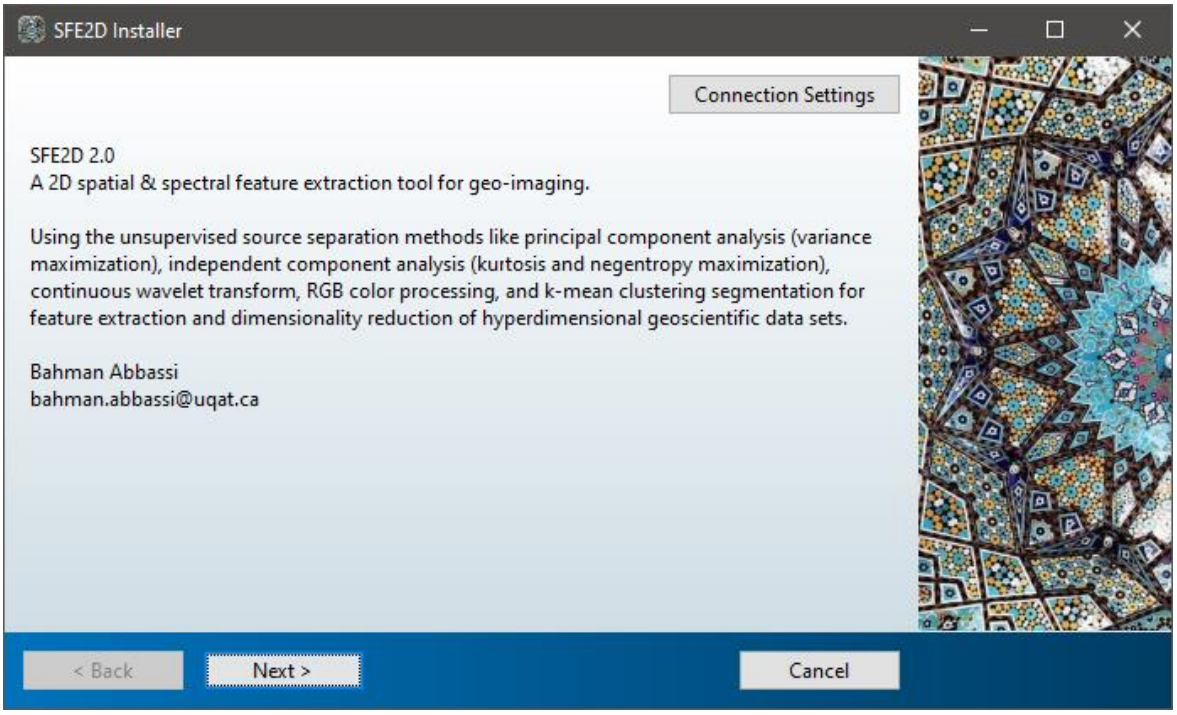


Figure 2

Press next to define the installation location for extracted files:

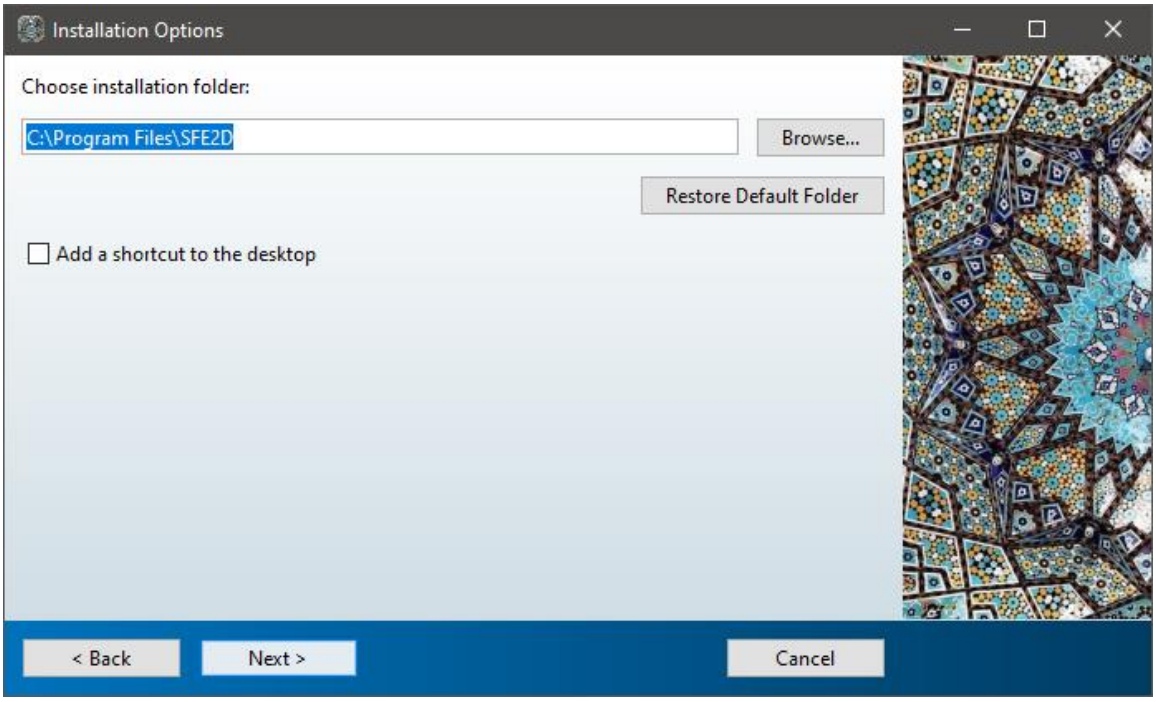


Figure 3

And then the location of Matlab Runtime files:

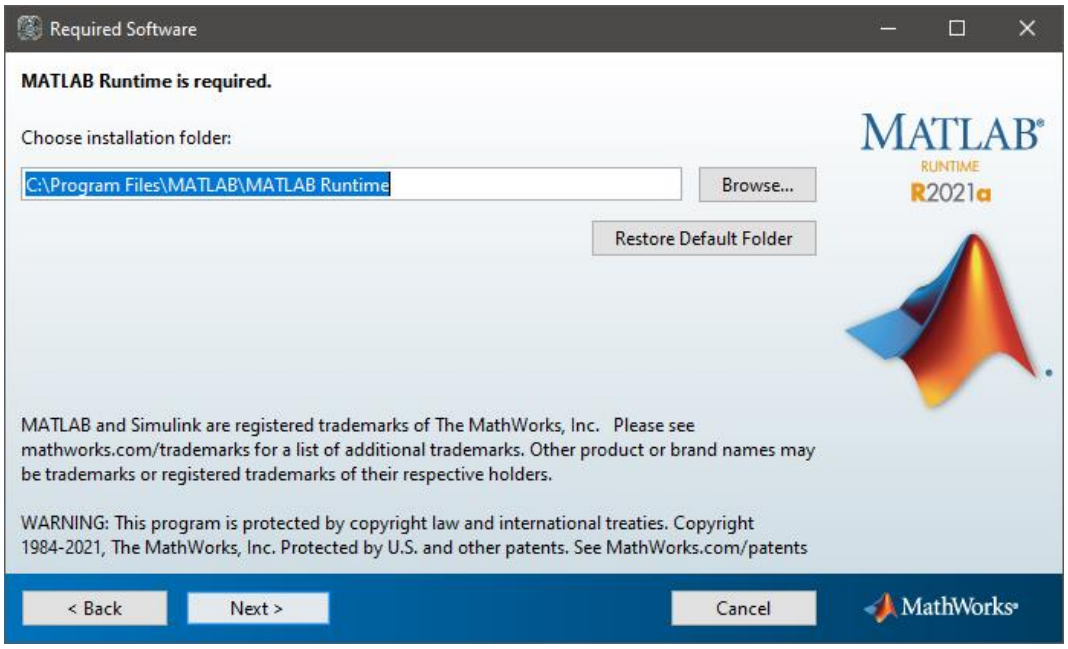


Figure 4

After accepting the terms of the license agreements, pressing “Install” should finalize the installation process:

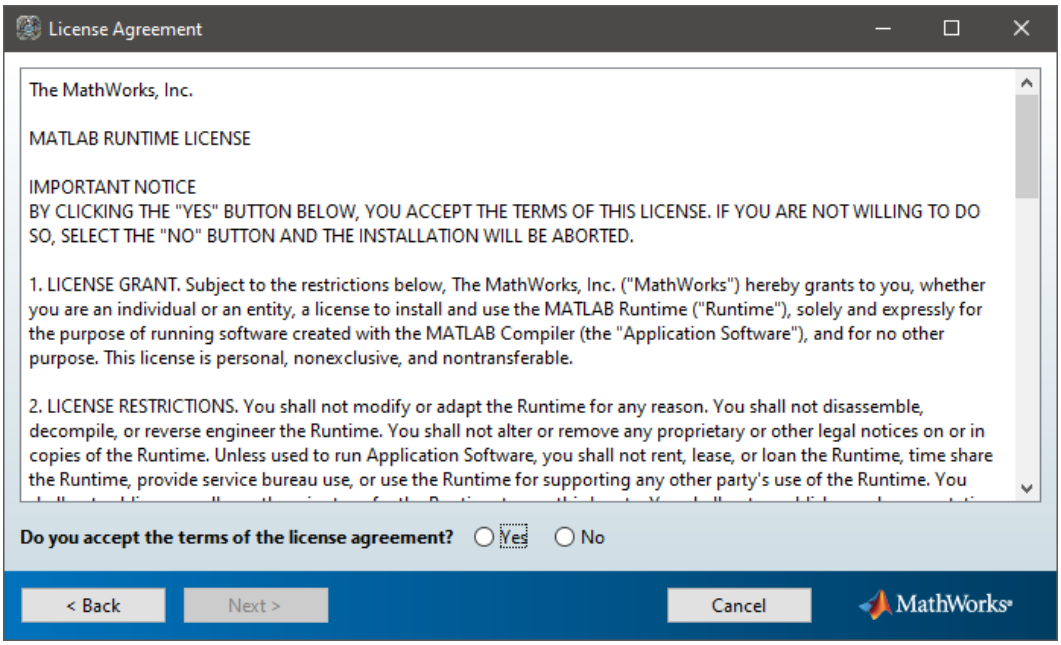


Figure 5

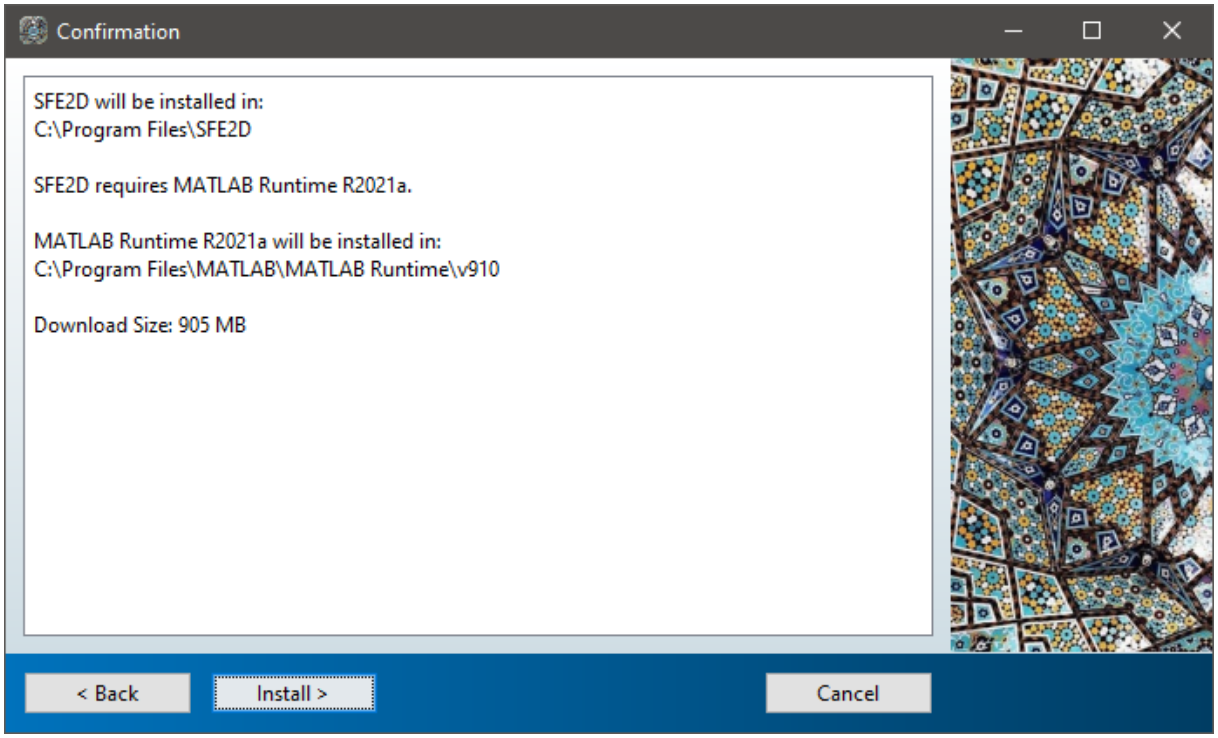


Figure 6

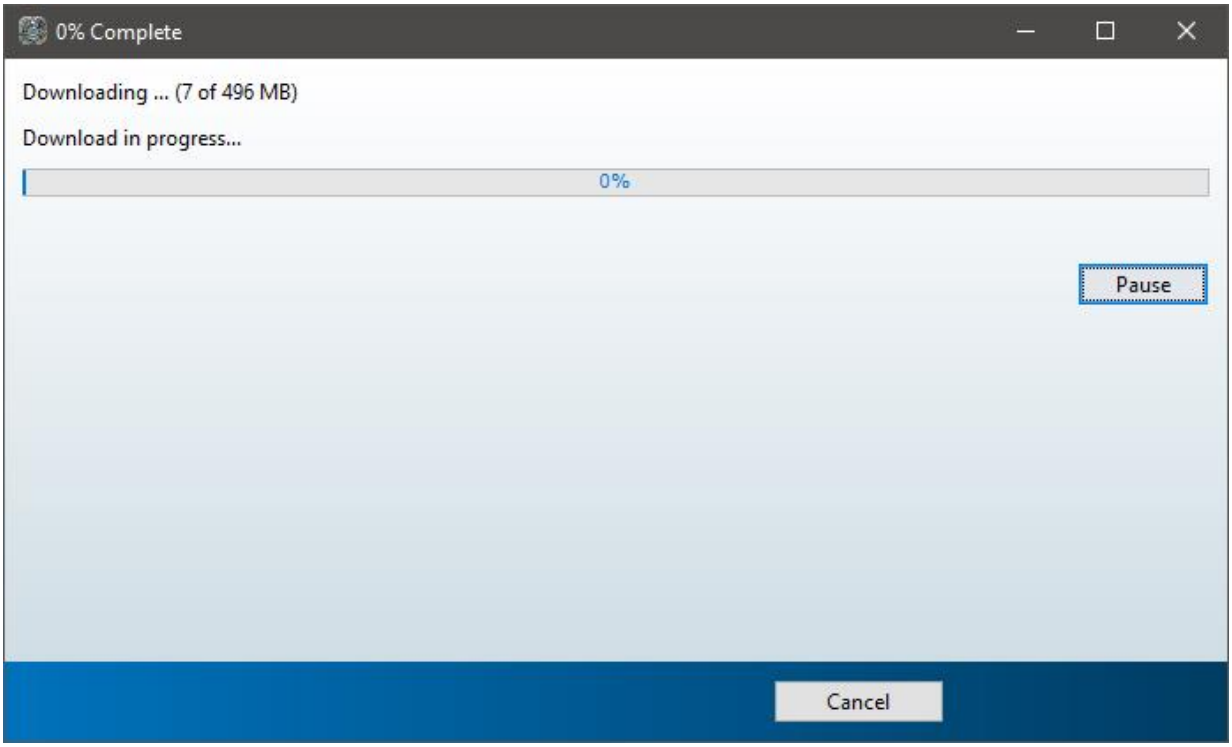


Figure 7

Users can also directly download the runtime from the address presented below:

https://ssd.mathworks.com/supportfiles/downloads/R2019b/Release/5/deployment_files/installer/complete/win64/MATLAB_Runtime_R2019b_Update_5_win64.zip

After unzipping the file and installing the runtime, the program should work by simply double click on the SFE2D.exe. It is recommended to copy/paste the SFE2D.exe to a project folder where data sets are located with read/write permission.

Running the program for the first time needs an activation. To activate the program, one needs to double click on the *Activate.exe* file to create a *Key* file. A dialogue box (*Password Required*) appears that asks to insert the password provided for the user. As soon as typing it and pressing Enter, a *Key* file is going to be generated. The user needs to keep that file and save it in the same project folder.

When running the SFE2D, the program interface appears as below:

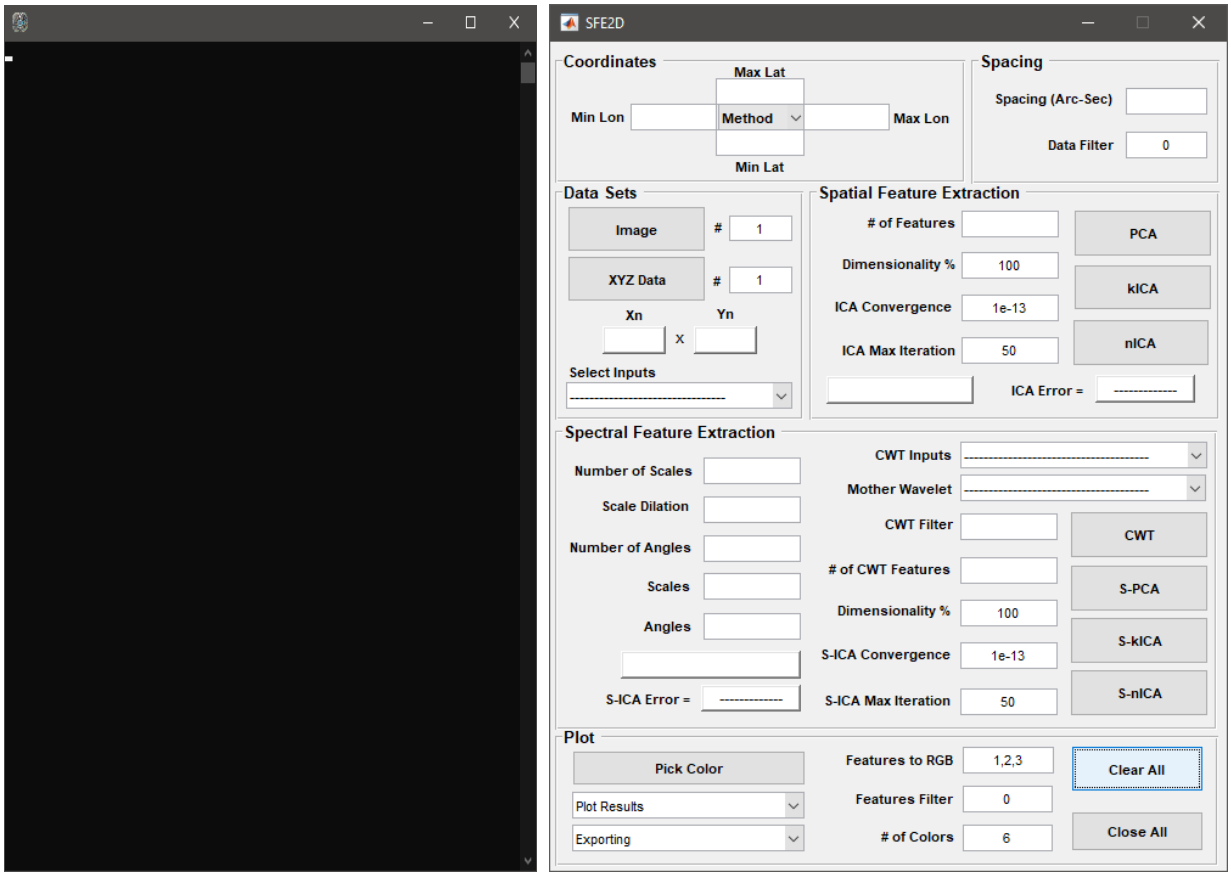


Figure 7

On the right side, the user can control the program parameters, and on the left, the progress of calculations and possible errors can be tracked with the windows command shell console for execution (black screen on the left).

3. Copyright

SFE2D is copyrighted by © B. Abbassi, 2020. All rights reserved. No part of this document or the related files may be reproduced or transmitted in any form, by any means (electronic, photocopying, recording, or otherwise) thereafter without written permission from the author. Contents are subject to change without notice.

4. Theory

Multivariate feature extraction is an important topic in the automated interpretation of multiple geoscientific images (geo-images). In exploration geophysics, treatment of multiple sources of data put forth a profound pattern recognition problem that demands extraction of relevant features from multiple sources of geoscientific data sets in the form of numerous 2D/3D geo-images. However, in most cases, complex "shadow effects" add heterogeneous overprints on the geo-images and imposes difficulties in identifying the latent features. Sources of these shadow effects are numerous, including vegetation, water, snow, urbanization, clouds, pollutions, and rocks' alteration. These overprints are more or less present in most geoscientific data sets, and some of them are treated as noise while others are considered as valuable targets depending on the exploration purpose.

Basically, the challenge can be seen as a "hyperdimensional overlapping" problem. Several techniques are available for feature extraction in the high dimensional space. The question is how to detect the latent features from large sets of hyperdimensional geo-images (Abbassi, 2018). The PCA and ICA algorithms (Hyvärinen and Oja, 2000) are useful for spatial source separation and dimensionality reduction. However, some images include unique features that are not seen on any other geo-images. In this case, spectral decomposition provides a unique way for feature extraction in the frequency domain.

In this program, a 2D feature extraction scheme is proposed that combines 2D CWT with variance, kurtosis, and negentropy maximization algorithms (PCA, kICA, and nICA) to extract spatial/spectral features from multiple images. The algorithm consists of three main stages (Figure 8):

1. ***Preprocessing***: The 2D interpolation and filtering of raw datasets to prepare the input images for feature extraction.
2. ***Spatial Feature Extraction***: A preliminary PCA/ICA as spatial source separation and dimensionality reduction.
3. ***Spectral Feature Extraction***: That consists of two substages:
    a. ***Continuous Wavelet Transform*** (CWT).
    b. ***Spectral PCA/ICA*** (SPCA/SkICA/SnICA).

The CWT (stage 3.a) can also be performed directly on the input images without any spatial feature extraction. It can also be used on a single image to analyze its spectral properties. Different mother wavelet algorithms are integrated into the SFE2D program with different isotropic and anisotropic properties, including:

- *Gaussian (Asymmetric)*
- *Difference-of-Gaussian (Symmetric)*
- *Mexican Hat (Symmetric)*
- *Morlet (Asymmetric)*
- *Isotropic Morlet (Symmetric)*
- *Cauchy (Asymmetric)*
- *ES-Cauchy (Asymmetric)*
- *Paul (Symmetric)*
- *Wheel (Symmetric)*
- *Pethat (Symmetric)*
- *End-Stop (Asymmetric)*
- *Gabor Mexican Hat (Asymmetric)*
- *Sinc (Symmetric)*
- *Fan (Symmetric)*

This tool can separate statistically overlapped features without specific learning examples. The outputs here are maximally separated features from geo-images. Geophysical methods often show highly overlapped features on multiple data sets. The independent components produced with this tool not only makes the overlapped features visible but also removes the recurrent patterns through dimensionality reduction. This helps us to summarize the unique differences between multiple geo-images rather than the recurring similarities. This procedure makes interpretations much easier for human users as well as further supervised deep learning methods like feature selection.

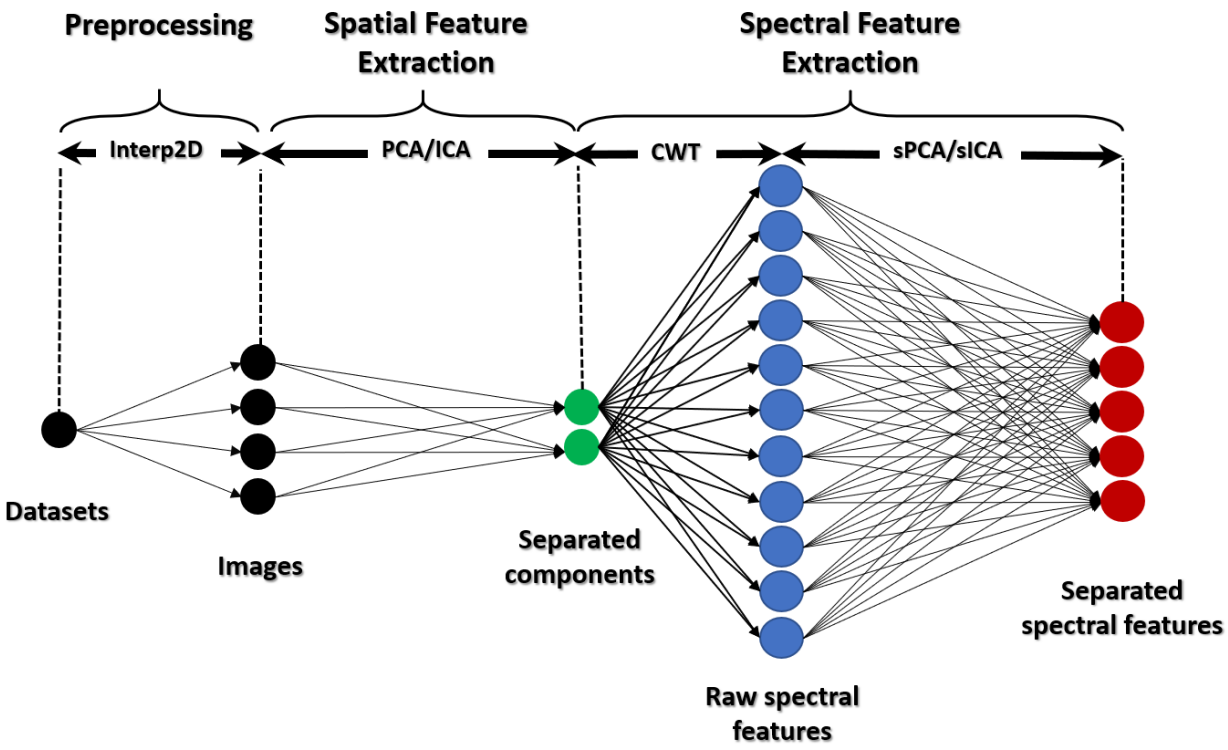


Figure 8: Schematic view of the SFE2D procedure.

5. Input/output formats

After defining the minimums and maximums of the geographic coordinate system (latitude and longitude in decimal degrees) and the spacing (latitude and longitude in Arc-Second), the program supports two groups of inputs: RGB color images and point data sets. The color images can be read in the form of BMP/PNG/JPG formats. Depending on the size of the project (minimum/maximum of coordinates and spacing), the number of pixels is automatically adjusted. The point data sets should be in the form of Comma-Separated Values (CSV) with three columns. The first column should be the longitude values (X), the second column should be the latitude values (Y), and the third column should be the measured/observed values (Z).

The output of the program can be in many forms:

- RGBs of separated components. Users have control over the selection of the desired components to compile their models.
- Interactive selection of the region of interest (ROI) based on RGB colors.
- Color segmented maps for zoning of different extracted features by a fast k-means clustering algorithm.
- All matrices as CSV and GRD files (Golden Software Surfer, ASCII format).

6. Program environment

The SFE2D program is designed for spatial and spectral feature extraction of an unlimited number of geo-images. The program is made of six main windows, including:

- **Coordinates**: Preferably, the geographic coordinate system in SFE2D v.1.4.

Two methods are used for clipping the data sets:

A) Rectangular coordinates:

Coordinates are readable from a prepared text file with this format:

First line:	Min Longitude (Min Lon)
Second line:	Max Longitude (Max Lon)
Third line:	Min Latitude (Min Lat)
Fourth line:	Max Latitude (Max Lat)

For example, in the case above:

-77.8

-77.2

53.15

53.45

For satellite images, a separate text file is also needed to upload the original image coordinates for the purpose of georeferencing.

If you use microscopic images without georeferencing, you just use a text file with this content if the image has a square size (x and y pixels are the same):

0

1

0

1

If not, you need to consider the number of pixels in each direction. For example, if your image has a 1138 x 866 resolution, you need to use a ratio compatible with this resolution. The coordinate text file will look like this:

0

1.314

0

1

Where the 1.314 is the ratio of horizontal to vertical resolution of input image (1138 /866). When you upload the image, you need to read the same text file again.

B) Automatic polygonization:

Sometimes geophysical data sets are geometrically irregular, and they will not cover any rectangular area. For these cases, program automatically polygonise the 2D space based on the irregular coordinates. You can also digitize the region of your interest on any GIS software and use the XY coordinates for automatic polygonization in SFE2D. You just simply need to upload the CSV file related to that data sets with at least two X and Y coordinates columns.

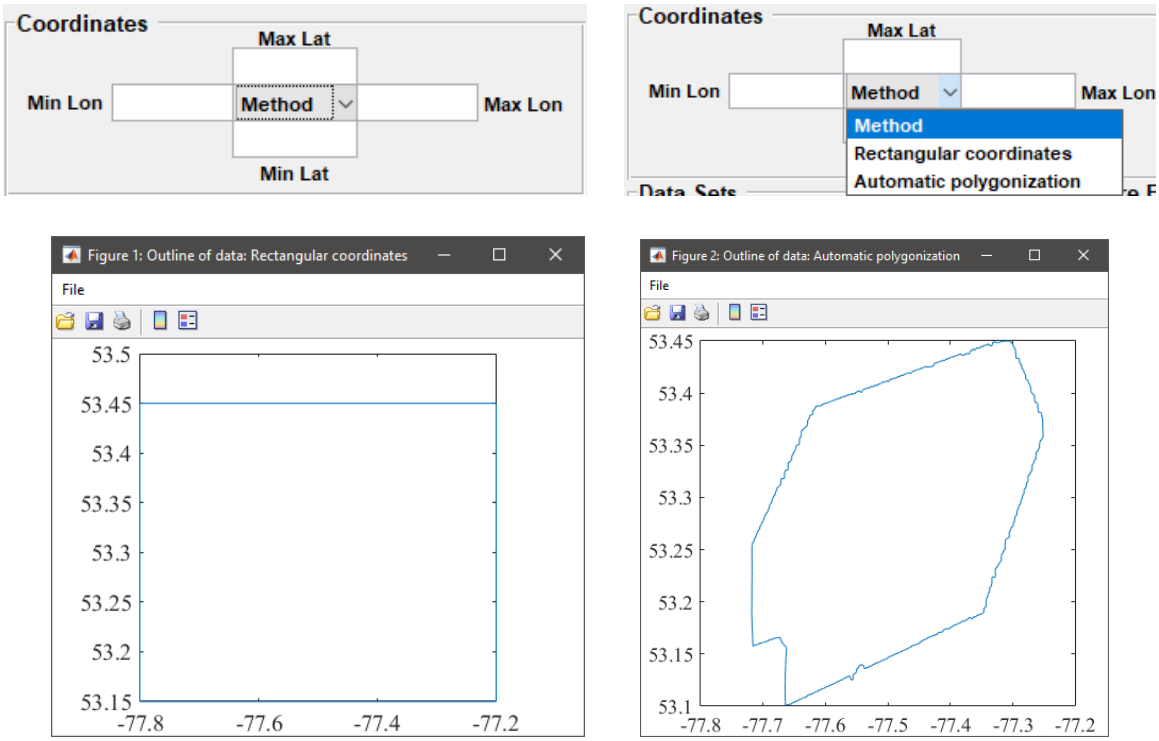


Figure 9

- **Spacing**: X and Y (Longitude and Latitude) spacings are in arcsecond. On average, in Quebec, each Arc-Sec of latitude is about 33 meters, and each Arc-Sec of longitude is about 17 meters. The ratios are automatically adjusted for any location on earth so that the overall scales of the images are affected for optimal presentation.

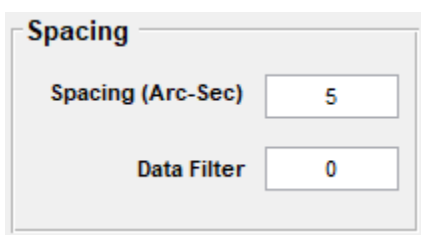


Figure 10

- **Data sets**: To import RGB satellite images as well as XYZ format of point data sets (magnetic, gravity, elevation, gamma-ray concentrations, and so on). If needed, we can add support for hyperspectral imageries in the next versions.

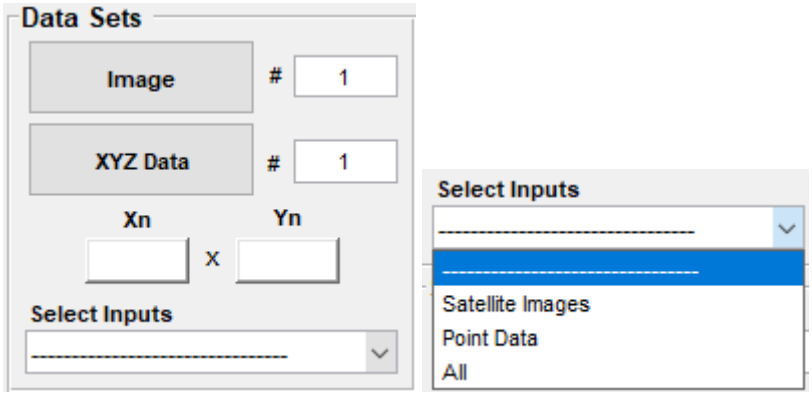


Figure 11

- **Spatial Feature Extraction**: For spatial feature extraction of multiple geo-images through PCA, kICA, and nICA. The number of features has to be equal to or less than the number of input geo-images (in the case of dimensionality reduction). Convergence and maximum iteration of the ICA methods are also available for users to change in the full version of the software. The demo version of the software is limited to 10 iterations and a threshold of 0.001 for convergence criteria.

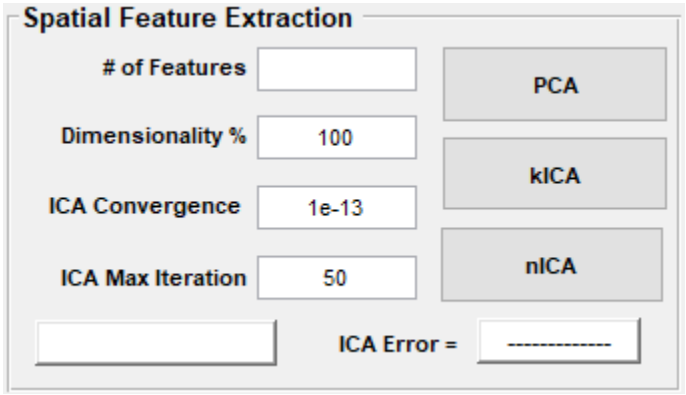
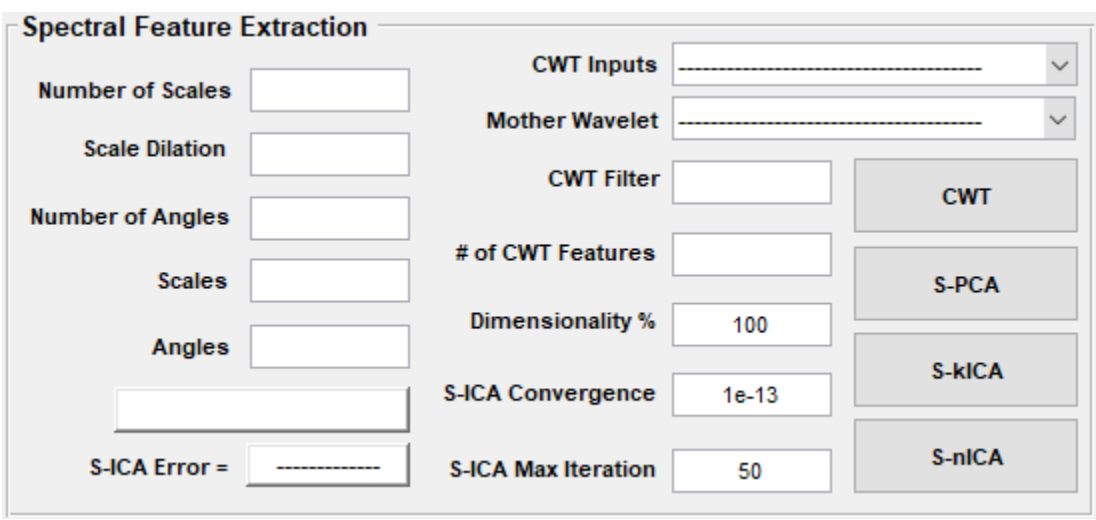


Figure 12

- **Spectral Feature Extraction**: This window consists of two major sections. Firstly, a 2D CWT to decompose desired inputs into raw spectral features. Users have control over the number of scales, scale dilation, and the number of angles in which the mother wavelet will surf. Then, the scales vector, corresponding frequencies, and angles vector are automatically calculated for each mother wavelet structure. Various isotropic and anisotropic mother wavelets are also presented here to be used by the user based on the purpose of the studies. Secondly, S-PCA, S-kICA, and S-nICA are also available for spectral source separation. After CWT, the program automatically sets the number of CWT features. Users need to decide whether dimensionality reduction is required or not (e.g., 100% for no dimensionality reduction). For spectral ICA methods (S-kICA and S-nICA), convergence and maximum iteration criteria are also needed. The demo version of the software is limited to 10 iterations and a threshold of 0.001 for convergence criteria.



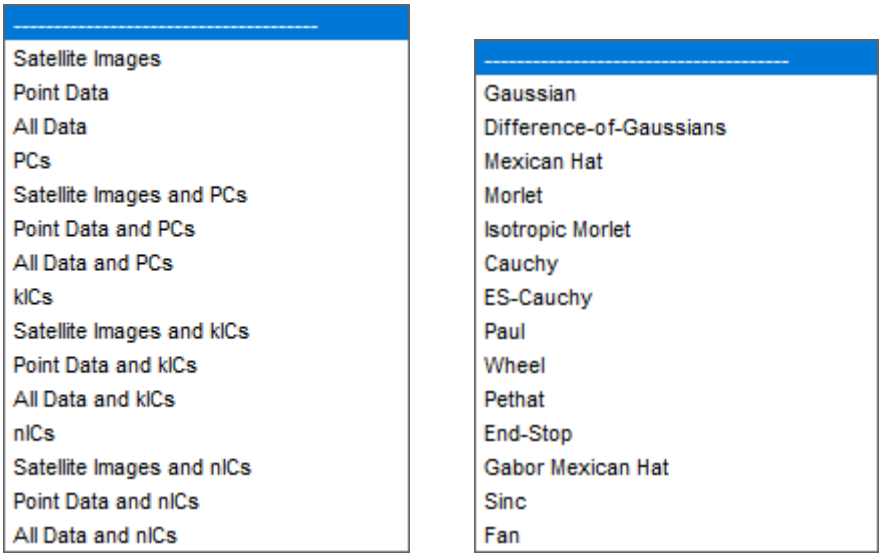


Figure 13

- **Plot**: This window is designed to illustrate and export the data sets and results. The program enables the user to pick the desired extracted features and integrate them in red/green/blue (RGB) images for efficient visualization of underlying features. Users can also export results in XYZ formats and grid formats (.GRD) compatible with the Golden Software Surfer program.

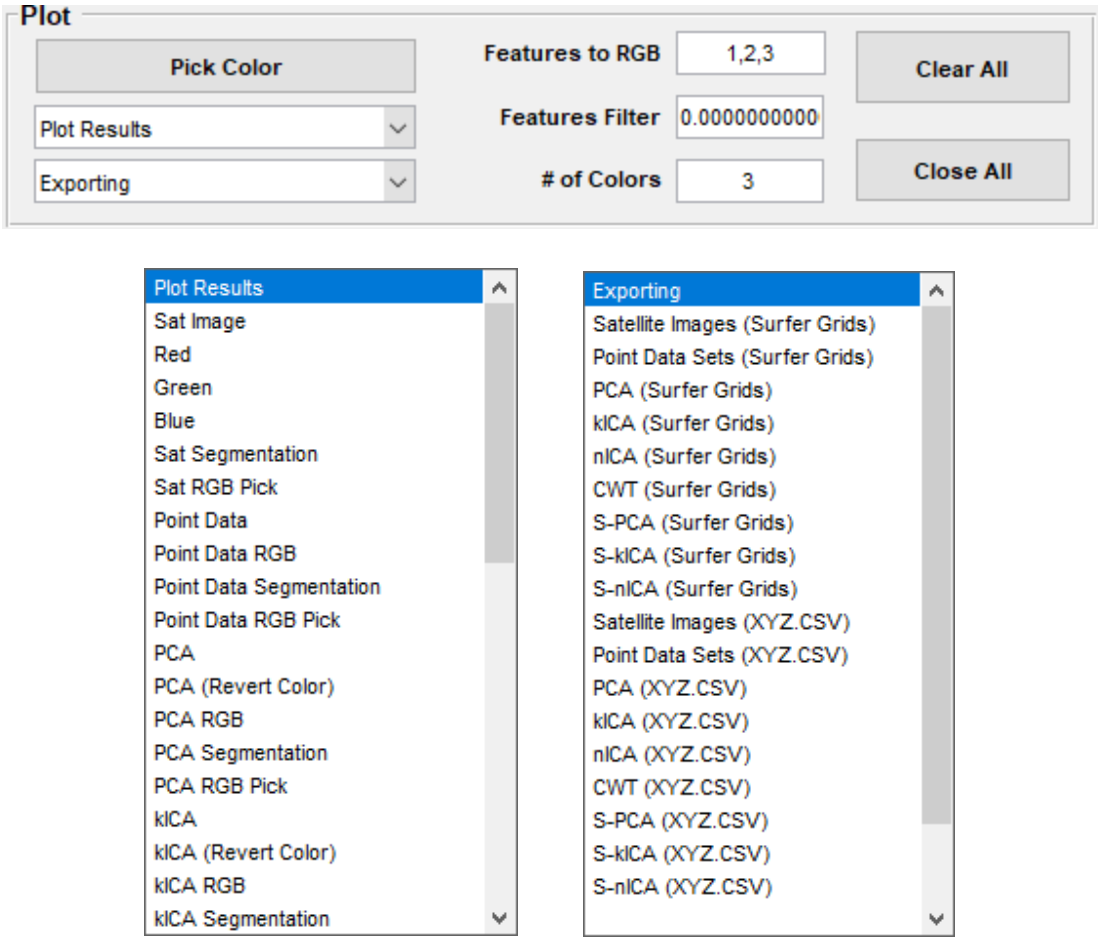

Plot
Pick Color
Plot Results
Exporting
Features to RGB
1,2,3
Features Filter
0.0000000000
# of Colors
3
Clear All
Close All
Plot Results
Sat Image
Red
Green
Blue
Sat Segmentation
Sat RGB Pick
Point Data
Point Data RGB
Point Data Segmentation
Point Data RGB Pick
PCA
PCA (Revert Color)
PCA RGB
PCA Segmentation
PCA RGB Pick
kICA
kICA (Revert Color)
kICA RGB
kICA Segmentation
Exporting
Satellite Images (Surfer Grids)
Point Data Sets (Surfer Grids)
PCA (Surfer Grids)
kICA (Surfer Grids)
nICA (Surfer Grids)
CWT (Surfer Grids)
S-PCA (Surfer Grids)
S-kICA (Surfer Grids)
S-nICA (Surfer Grids)
Satellite Images (XYZ.CSV)
Point Data Sets (XYZ.CSV)
PCA (XYZ.CSV)
kICA (XYZ.CSV)
nICA (XYZ.CSV)
CWT (XYZ.CSV)
S-PCA (XYZ.CSV)
S-kICA (XYZ.CSV)
S-nICA (XYZ.CSV)

Figure 14

7. Spatial feature extraction with SFE2D

Spatial feature extraction refers to the application of blind source separation methods in the spatial domain to extract the previously mixed features from multiple geo-images. Dimensionality reduction is also possible. A standard way to perform a spatial feature extraction in SFE2D is as the following workflow:

- Read the coordinates.

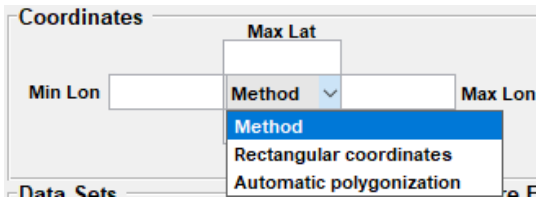


Figure 15

- Set Spacing and Data Filter values. If you do not want to smooth the data sets, just insert zero in the box any value greater than 0 will smooth the data sets proportional to the magnitude of the Data Filter.

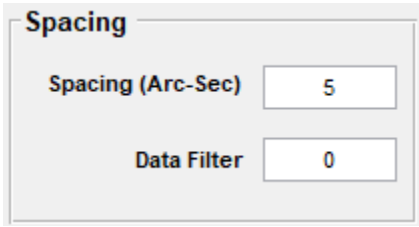


Figure 16

- Push *Image/XYZ Data* buttons in the *Data Sets* module and select the desired data sets.

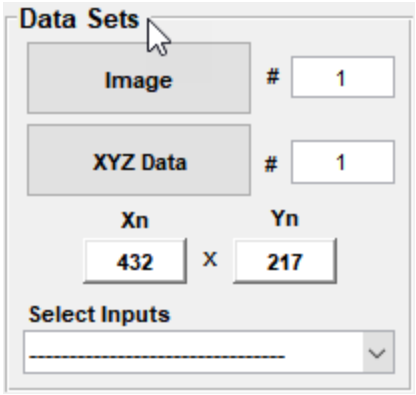


Figure 17

- To insert more images or XYZ data, change the value in the # box on the right to 2, 3, 4, and so on. For example, for the second image, the # box is 2, and for the third image, it would be 3.

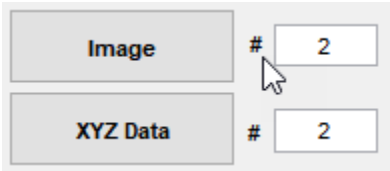


Figure 18

- The number of pixels in X and Y directions are also updated in Xn and Yn boxes.

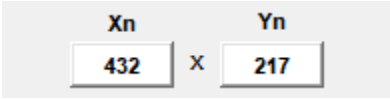


Figure 19

- The next step is to select the data sources for feature extraction (Select Input box). One can either take Satellite Images, Point Data, or All of them together. As soon as selecting the desired data sets, the # of features box is updated in the Spatial Feature Extraction module.

Figure 20

- Users can reduce the dimensionality by changing the value of the Dimensionality % box. 100% means no dimensionality reduction; that is, for 100 features, the output of feature extraction is the same.

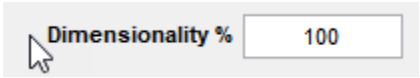


Figure 21

- ICA methods also need two other parameters: ICA Convergence and ICA Max Iteration. ICA Convergence, by default, is set to 1e-13. If faster results are needed, you can reduce it to 1e-6 to 1e-5 at the expense of lesser source separation accuracy. One can also increase the number of ICA maximum iteration for better convergence to eliminate the local minima traps during optimization. By default, the number of Max Iterations is set to 50 iterations.

Figure 22

- The next step is to run the PCA (variance maximization), kICA (kurtosis maximization), and nICA (negentropy maximization) algorithms.

Figure 23

- After each run, a message appears that ensures the algorithm is successfully deployed (*PCA Completed, kICA Completed, nICA Completed*).

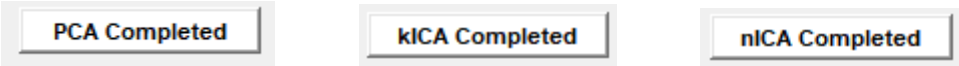


Figure 24

- For kICA and nICA methods, the best convergence error is also presented in a small box ICA Error.

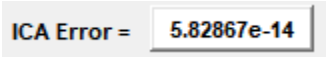


Figure 25

- A Figure window also automatically pops up that shows the ICA algorithms' performances both in linear and logarithmic plots. The logarithmic scale helps users follow the fluctuations of optimization in small ICA errors that are not visible on a linear scale.

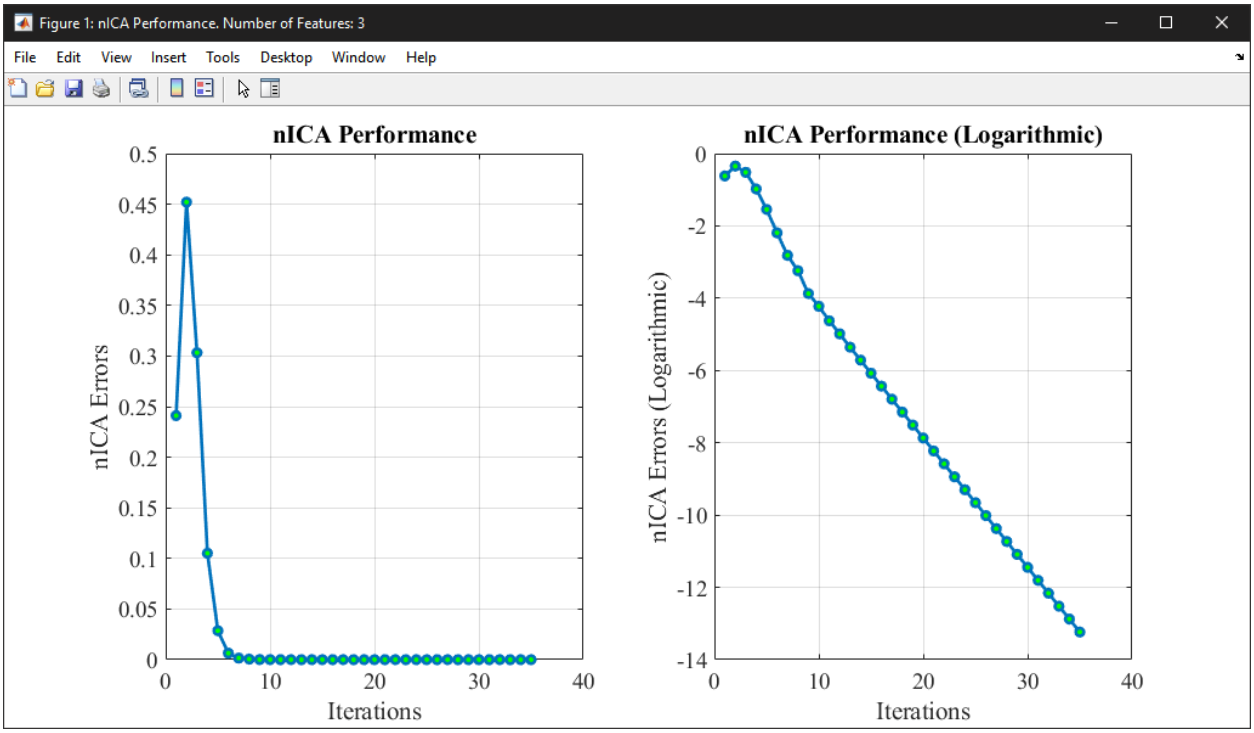


Figure 26

- Now we can see the results in the *Plot* module of the program (*Plot Results* box). We can display the input satellite images, separate RGB channels, and point data sets, together with PCA, kICA, and nICA results.

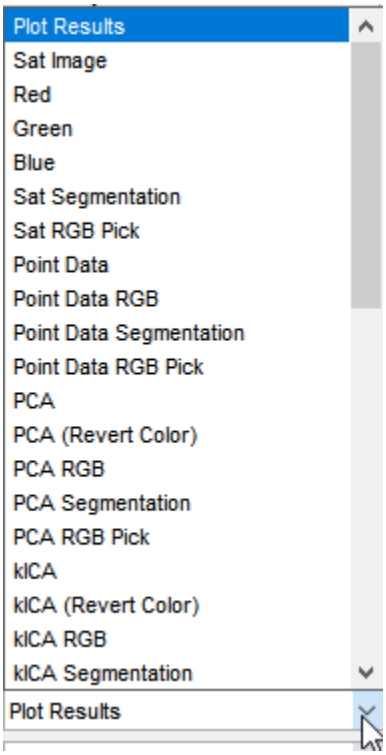


Figure 27

- For example, to display PCA results, we can select the *PCA* in the *Plot Results* box to visualize all the extracted features. *PCA (Revert Color)* shows the results with opposite polarities.

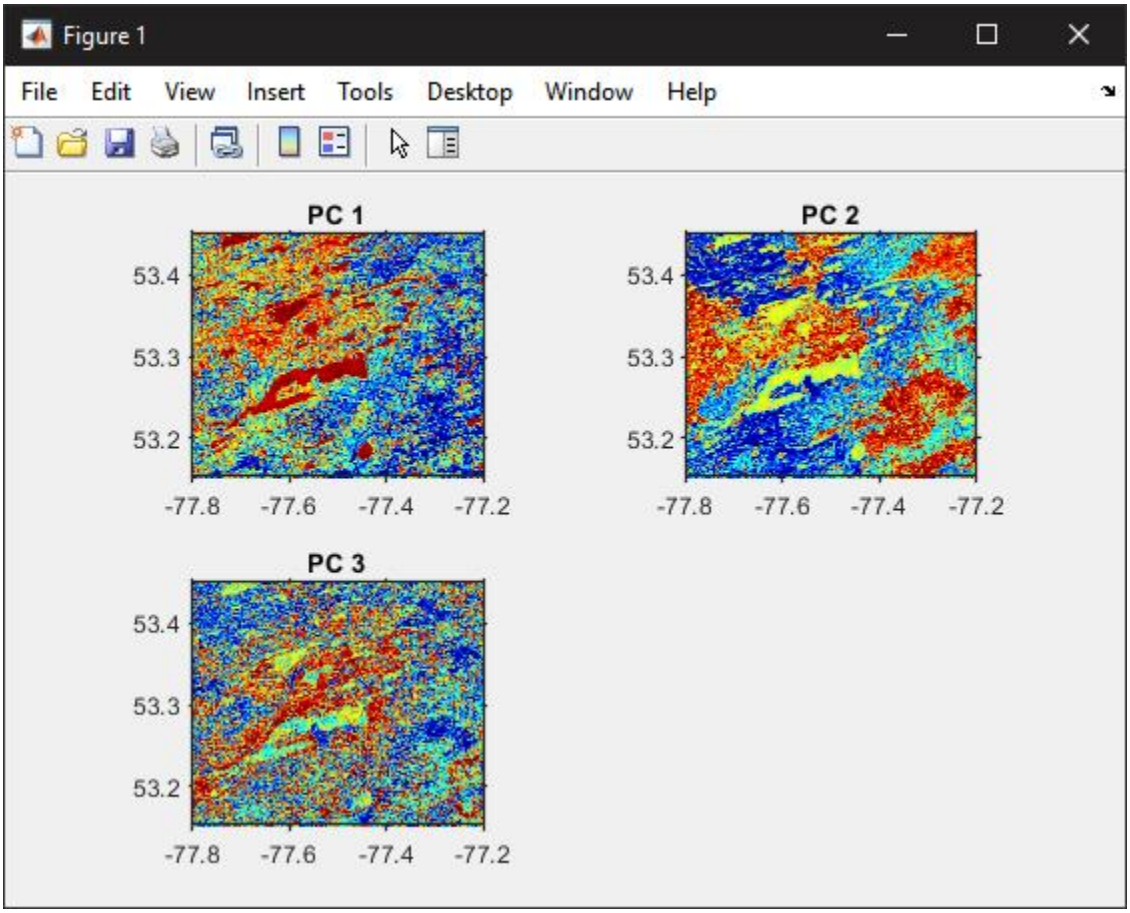


Figure 28

- We can also select the *PCA RGB* to compile a colored image out of three manually selected set of extracted features with the PCA method. To control which extract features are used for image compilation, you can set the label number of desired features in the *Features to RGB* box. For example, in nICA feature extraction, the 1, 2, 3 values mean the features 1, 2, and 3 are used for image compilation. You can also change the polarity of the features since PCA/ICA methods distort the polarities randomly. For example, you set -2, 1, 3 values in the *Features to RGB* box.

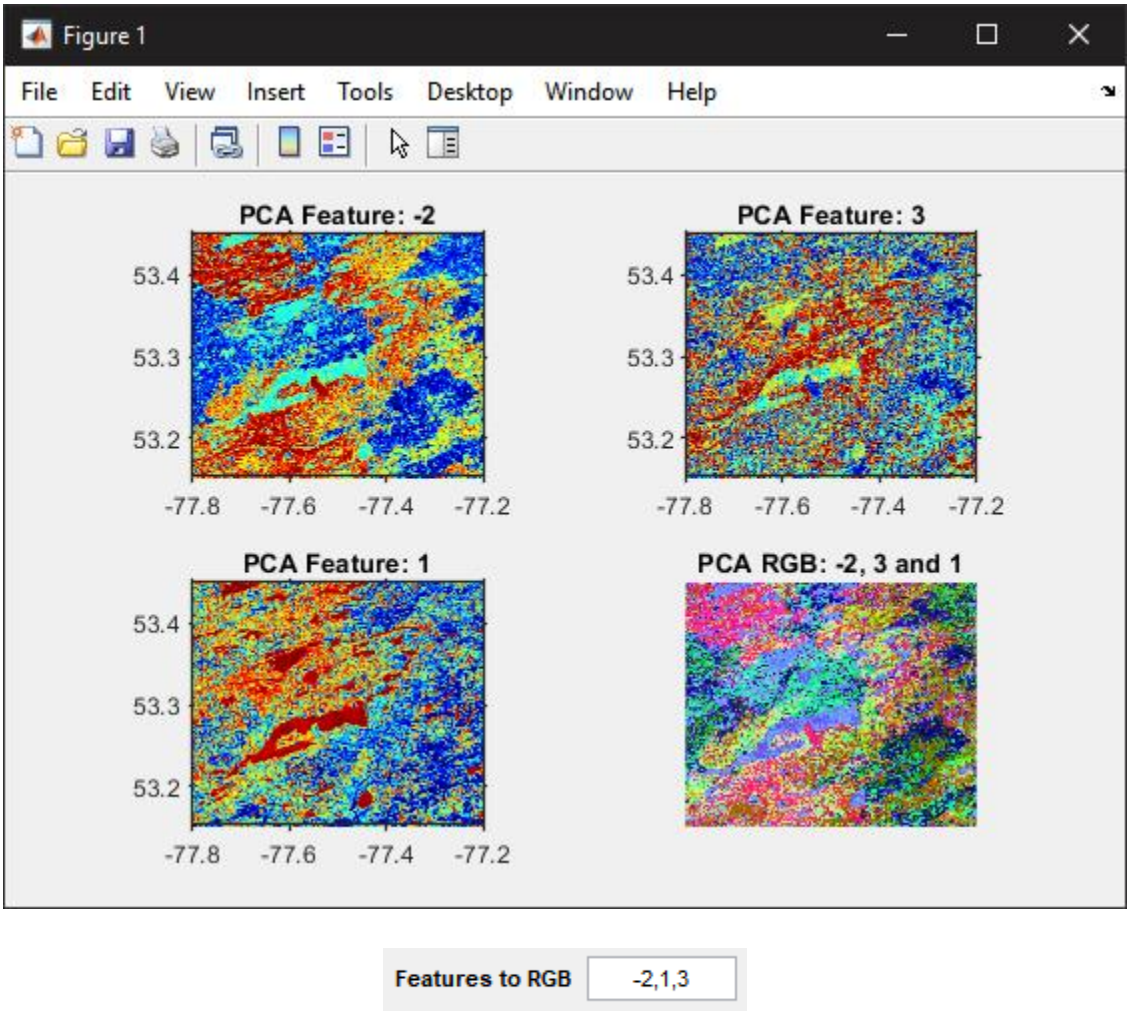


Figure 29

- *PCA/kICA/nICA RGB Pick* also displays the compiled image in a larger plot for better visualization.

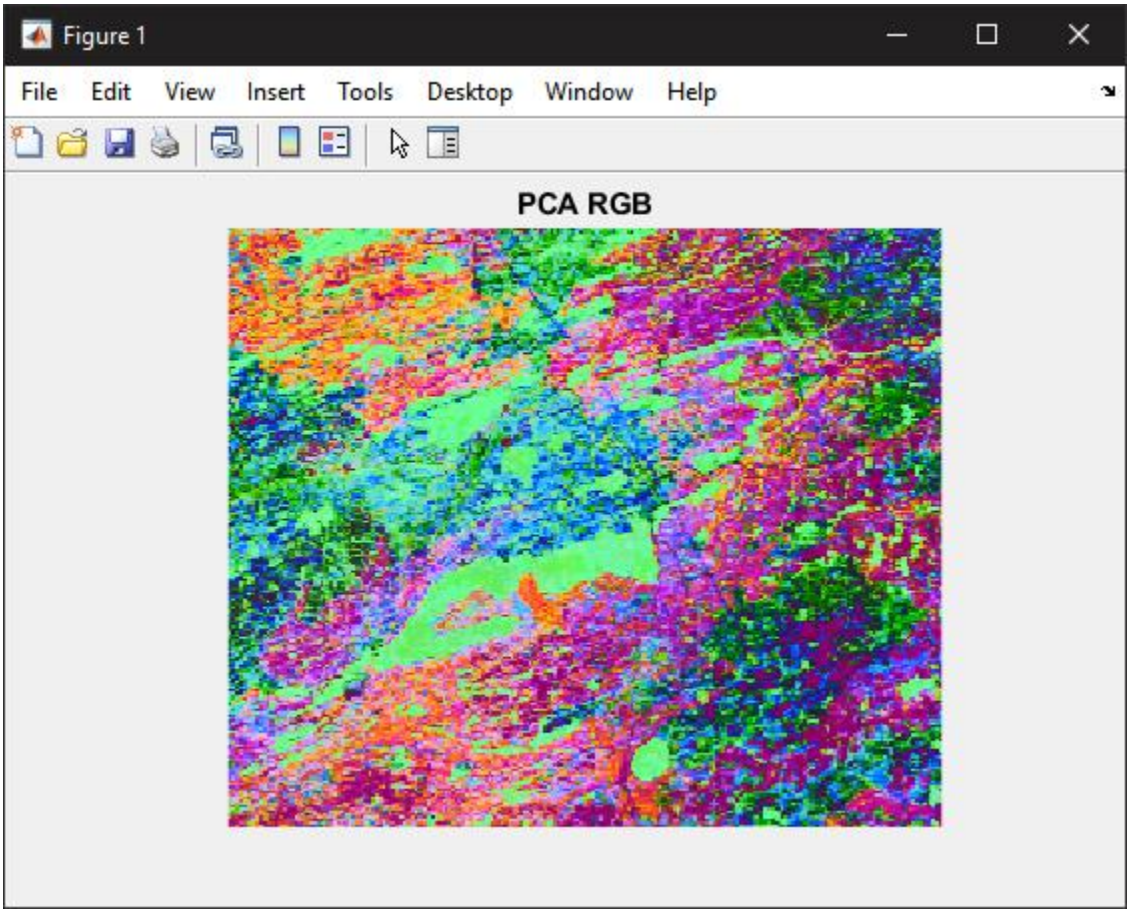


Figure 30

- For the application of *Pick Color* and Image Segmentation, please refer to sections 9 and 10.

8. Spectral feature extraction with SFE2D

Spectral feature extraction refers to the application of 2D continuous wavelet transform combine with source separation algorithms (PCA and ICA) to decompose and extract the frequency-dependent features. Dimensionality reduction is also possible. A standard way to perform a spectral feature extraction in SFE2D is as the following workflow:

- Read the coordinates.

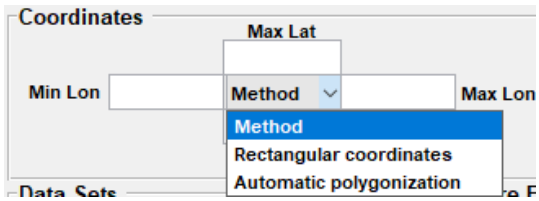


Figure 31

- Set Spacing and Data Filter values. If you do not want to smooth the data sets, just insert zero in the box any value greater than 0 will smooth the data sets proportional to the magnitude of the Data Filter.

Figure 32

- Push Image/XYZ Data buttons and select the desired data sets. Users can also apply the outputs from the spatial feature extraction stage as the input of the spectral feature extraction.

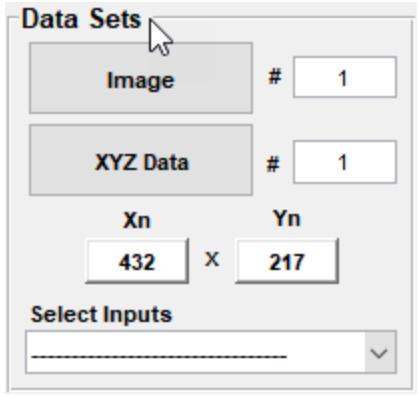


Figure 33

- In the *Spectral Feature Extraction* module, set the *Number of Scales*, *Scale Dilation*, and *Number of Angles*. For example, *Number of Scales = 3*, *Scale Dilation = 2*, and *Number of Angles = 8*, means that the mother wavelet expands three times at scales 1, 3, and 5 (scale dilation is the interval between scales) in 8 directions (45°, 90°, 135, 180°, 225°, 270°, 315°, 360°).

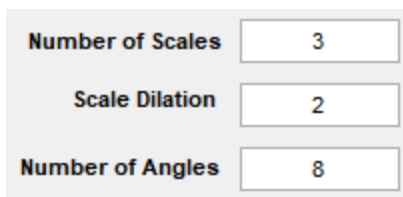


Figure 34

- Now select the desired inputs for spectral feature extraction in the CWT Inputs box. There is a list of possible options that allow users to pick a single data set, multiple images, or the PCA/ICA results from spatial feature extraction. As soon as selecting the desired CWT input, the program calculates the respective scales and angles for equal intervals and updates the *Scales* and *Angles* boxes. Users are now able to manually modify the scales and angles to suit their specific workflow.

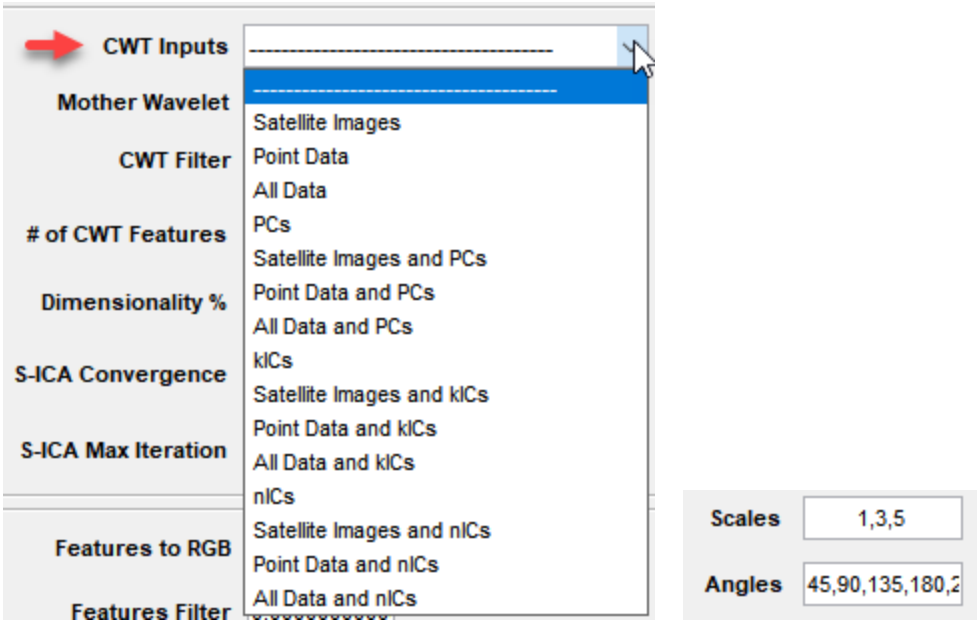


Figure 35

- Then, select the type of mother wavelet. A series of different symmetric/asymmetric mother wavelets are presented here. If you choose a symmetric mother wavelet, you should use a single direction for wavelet movement (*Number of Angles* = 1). You can modify the direction in the Angles box.

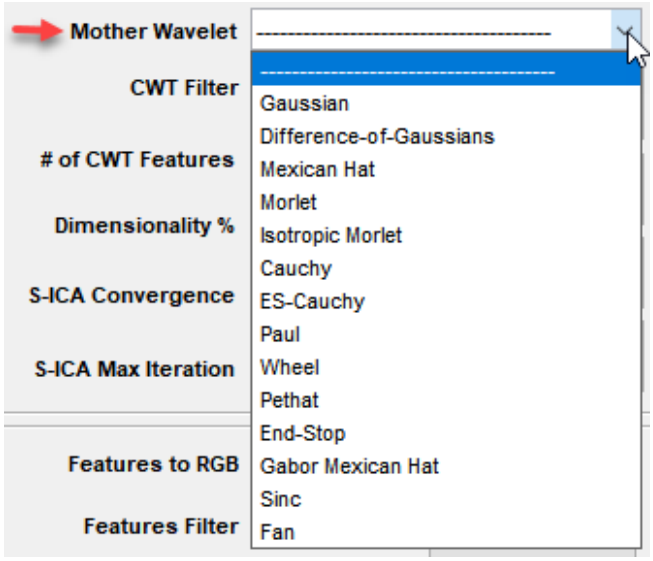


Figure 36

- CWT Filter box defines a way to set a damping factor for decomposed wavelet features. This damping factor helps to smooth the decomposed wavelet features as scales go up. Setting a very small value produces unwanted interpolation artifacts on the wavelet features. Very high *CWT Filter* values also eliminate valuable features. A handy rule is to set this parameter equal to the *Scale Dilation*.

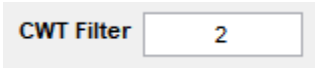


Figure 37

- The *# of CWT Features* is automatically calculated based on the number of scales and angles we set before.

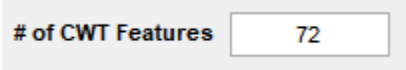


Figure 38

- We can also define how to reduce the dimensionality in the *Dimensionality %* box.

Figure 39

- Spectral ICA (S-ICA) methods need two other parameters: S-ICA Convergence and S-ICA Max Iteration. ICA Convergence, by default, is set to 1e-13. If faster results are needed, you can reduce it to 1e-6 to 1e-5 at the expense of lesser source separation accuracy. One can also increase the number of ICA maximum iteration for better convergence to eliminate the local minima traps during optimization. By default, the number of Max Iterations is set to 50 iterations.

Figure 40

- Now we can use a continuous wavelet transform algorithm (*CWT* button). Now, we can separate the decomposed wavelet features to reduce the frequency-dependent overlaps. The program uses PCA (variance maximization), kICA (kurtosis maximization), and nICA (negentropy maximization) algorithms for spectral feature extraction (S-PCA, S-kICA, and S-nICA).

Figure 41

- After each run, a message appears that ensures the algorithm is successfully deployed (*CWT Completed*, *PCA Completed*, *kICA Completed*, *nICA Completed*).

Figure 42

- For S-kICA and S-nICA methods, the best convergence error is also presented in a small box *ICA Error*.

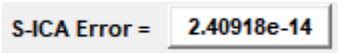


Figure 43

- A Figure window also automatically pops up, showing the ICA algorithms' performances both in linear and logarithmic plots. The logarithmic scale helps users follow the fluctuations of optimization in small ICA errors that are not visible on a linear scale.

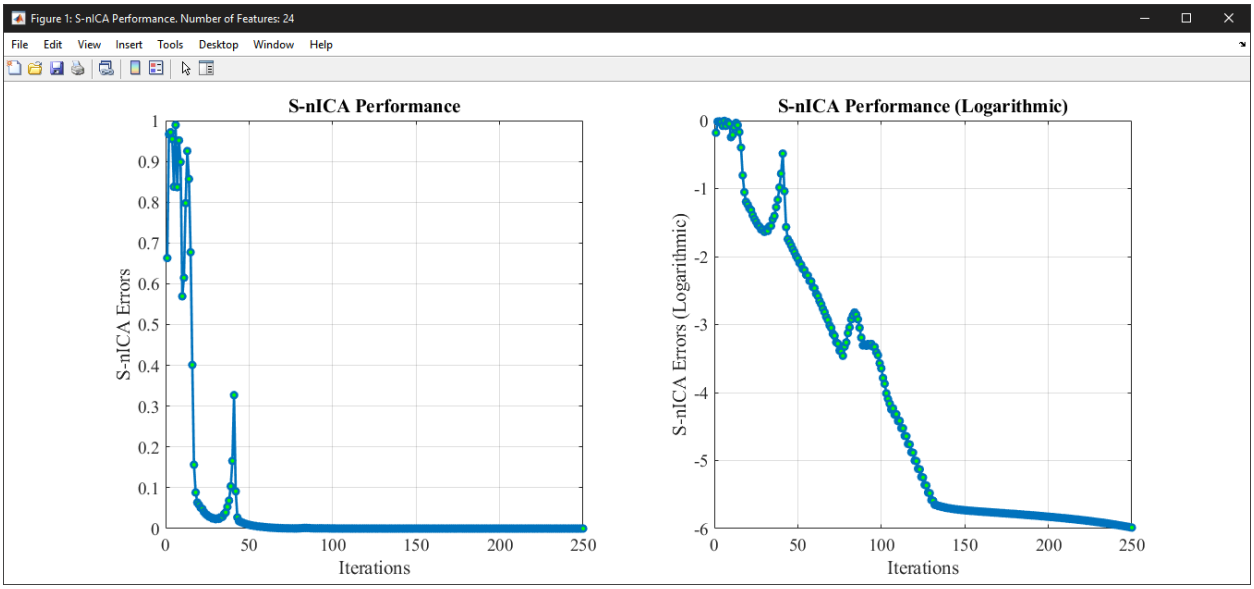


Figure 44

- Now we can see are results in the *Plot* module of the program in the *Plot Results* box. Here we can display the CWT, S-PCA, S-kICA and S-nICA results.

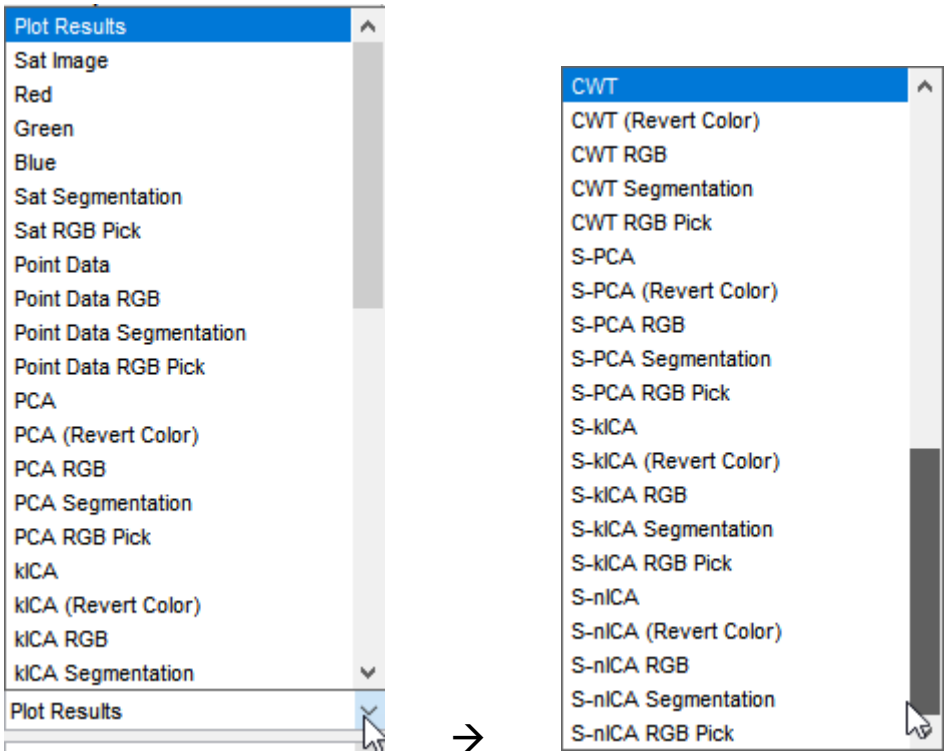


Figure 45

- For example, to display S-PCA results, we can select the *S-PCA* in the *Plot Results* box to visualize all the extracted features. We can also choose the *S-PCA RGB* to compile a colored image out of three manually selected set of extracted spectral features with the *S-PCA* method. To control which extracted features are used for image compilation, you can set the label number of desired features in the *Features to RGB* box. For example, in S-PCA feature extraction, the 1, 2, 3 values mean the features 1, 2, and 3 are used for image compilation. You can also change the polarity of the features since PCA/ICA methods distort the polarities randomly. For example, you can set -2, 1, 3 values in the *Features to RGB* box.
- *S-PCA/S-kICA/S-nICA RGB Pick* also displays the compiled image in a larger plot for better visualization.

- For the application of *Pick Color* and *Image Segmentation*, please refer to sections 9 and 10.

9. ROI selection with color pick algorithm.

A color pick algorithm is developed for the selection of the region of interest (ROI) based on color intensities. The program assigns the low and high thresholds for each RGB color band with several clicks on the screen. The more you click on the specified zones, the more accurate the thresholding works. Then, the program automatically picks up the minimums and maximums from red/green/blue bands. To reduce the effect of the outliers, only a range of 5 to 95 percentiles are kept.

A standard way to perform an ROI selection in SFE2D is as the following workflow:

- Display the RGB image by clicking on the options in *Plot* → *Plot Results* that end with the suffix *RGB Pick*.

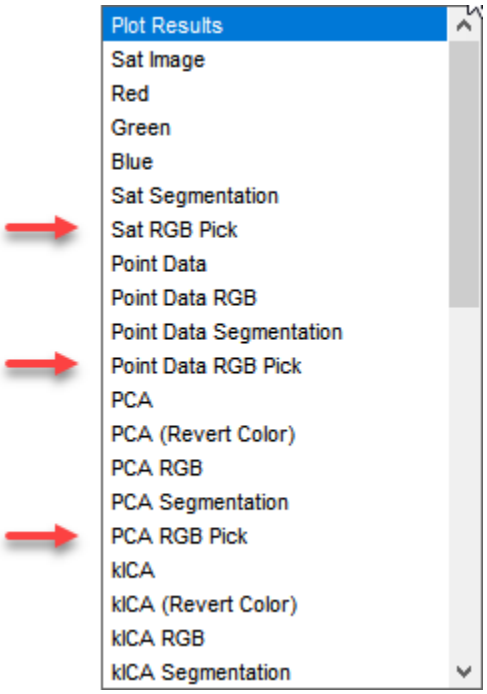


Figure 46

- Now in the *Plot module*, click on the *Pick Color* button.

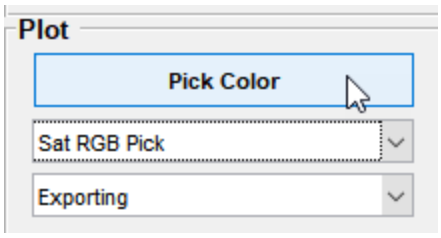


Figure 47

- Now repeatedly click on the places with interesting color signatures. The more you click, the more accurate the algorithm can detect the specified zone.

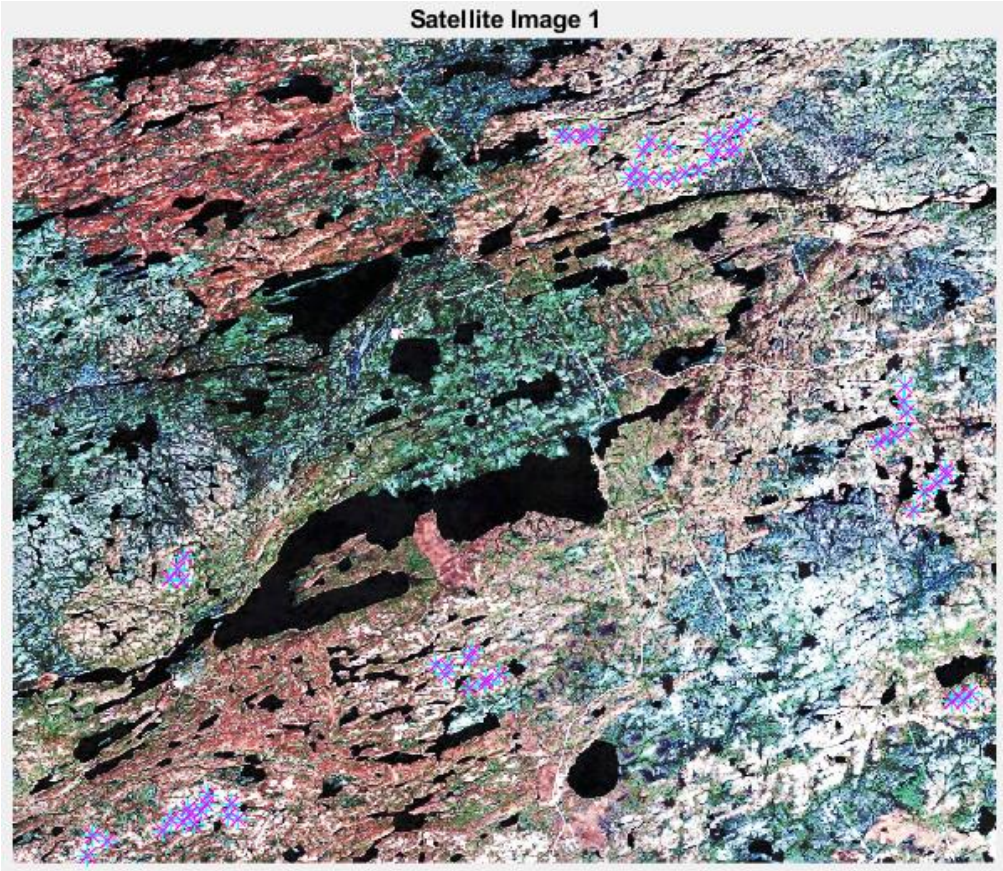


Figure 48

- After pressing the Enter, the program automatically finds all the places with similar color signatures like those you clicked and blanks out the rest.

Region of interest based on color range

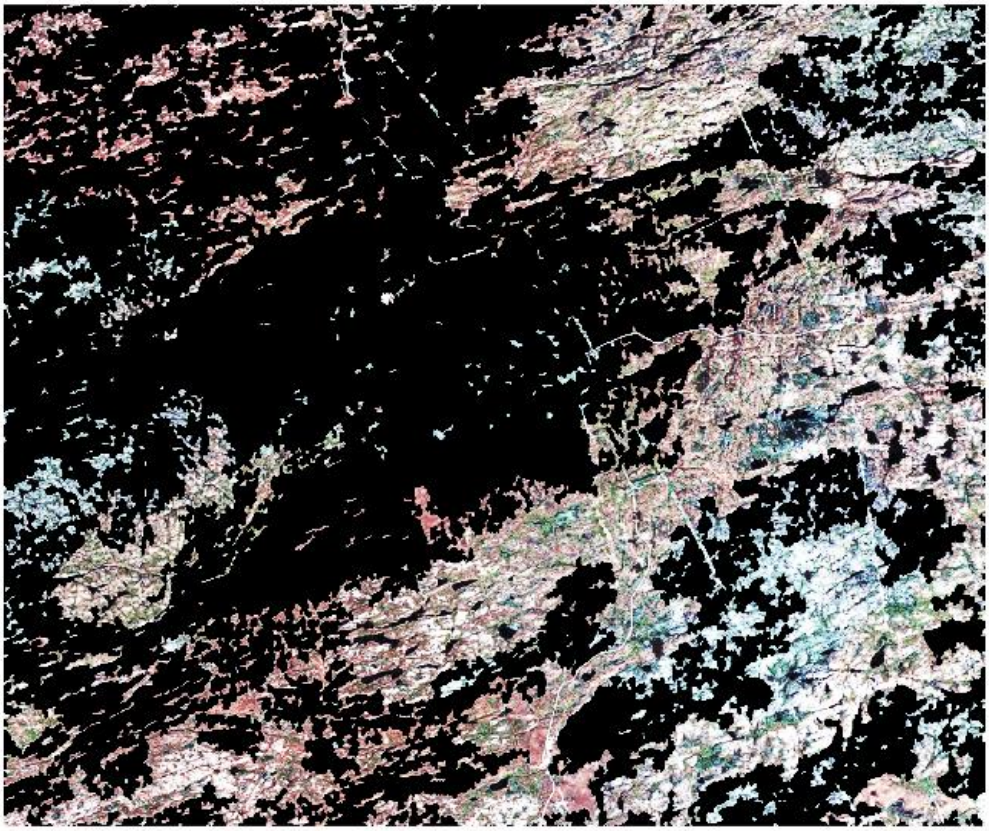


Figure 49

10. K-Mean clustering color image segmentation

SFE2D uses a fast k-mean clustering algorithm for the segmentation of color images. The segmentation helps to reduce the color space size to a manageable number of colors. The process produces a pseudo-geological map based on the spatial/spectral extracted RGB features, and eventually helps geologists detecting the hidden geological contacts and structures in geo-images. In addition, segmentation significantly reduces memory usage and speed up image analysis by focusing on relevant information.

The RGB space provides a more complex feature extraction problem where users can compile colored images from extracted features. Since every feature has two different polarities, we need to select three features out of $m = 2n$ features, where n is the number of extracted features. This is a permutation problem where $P(m,3) = m(m-1)(m-2)$ ways are possible to arrange three R/G/B features out of $m = 2n$ features. For three extracted features, we have six features, and we need to select three of them with different orders, which is 120 different ways. For four features, the number goes up to 336, and for five features, permutations are 720. As can be seen, increasing the dimensionality complicates how one can visualize feature extraction results in RGB space.

In the case of ICA-based segmentations in the RGB space, the problem gets more straightforward because the order of features is not a matter of concern in the *k-mean* segmentation. This is a combination problem where the number of ways to combine three features from a set of $m = 2n$ features is $C(m,3) = m(m-1)(m-2)/6$. However, this combination problem needs a few modifications because, although the order of features is not necessary, the order of polarities is essential in the calculation of 3D Euclidean distance in RGB space. Therefore, the problem can be reformulated in two parts:

1. Combinations of three features out of n extracted features: $C_1(n,3) = n(n-1)(n-2)/6$.
2. And the number of three combinations for all three R/G/B bands, which is $C_2 = \sum_{k=0}^{3} C(3,k) = 2^3 = 8$.

Therefore, the total number of combinations is $C_{Total} = C_1C_2 = 4n(n-1)(n-2)/3$. Table 1 shows all the possible permutations and combinations for up to 9 extracted features. As can be seen, the fraction of permutations to combinations (P/C) shows that the segmentation helps to reduce dimensionality between 15 times to 7.3 times. For very large dimensionalities ($n \rightarrow \infty$) it can be proved that the segmentation reduces the dimensionality down to six times. The SFE2D program displays all $C_2 = 8$ combinations for every C_1 combination. For very large dimensionalities, users can reduce dimensionality in ICA procedures to reduce the complexity of the problem.

Table 1. The total number of all RGB permutations (P) versus k-mean segmentations in RGB feature compilation.

n	***m***	***P***	C_1	C_2	***C***	***P/C***
3	6	120	1	8	8	15
4	8	336	4	8	32	10.5
5	10	720	10	8	80	9
6	12	1320	20	8	160	8.25
7	14	2184	35	8	280	7.8
8	16	3360	56	8	448	7.5
9	18	4896	84	8	672	7.28

A standard way to perform an ROI selection in SFE2D is as the following workflow:

- Set the # of Colors: For example, in this case, six colors are used for segmentation.

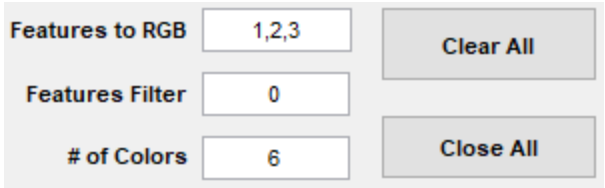


Figure 50

- Select Segmentation options in Plot → Plot Results.
- Select All Segments options in Plot → Plot Results. This option shows all eight possible combinations for the selection of three features. You can also plot all eight equivalent RGBs.

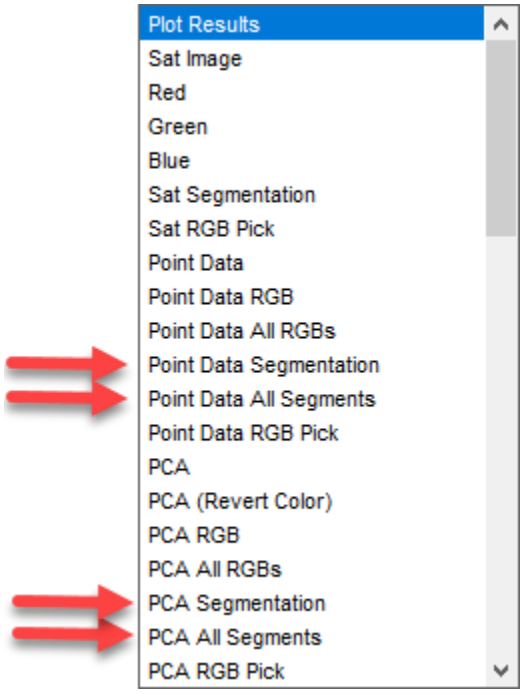
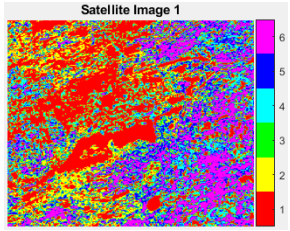
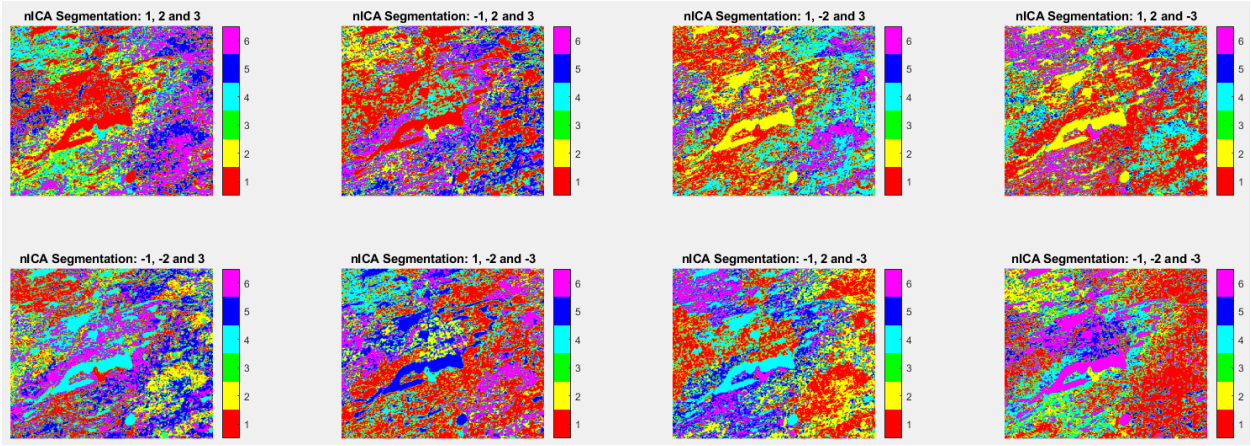


Figure 51

- An example of the segmentation on the Satellite image (from section 9) is presented below. As can be seen, spatial feature extraction has highlighted obscured structures in the segmented image.



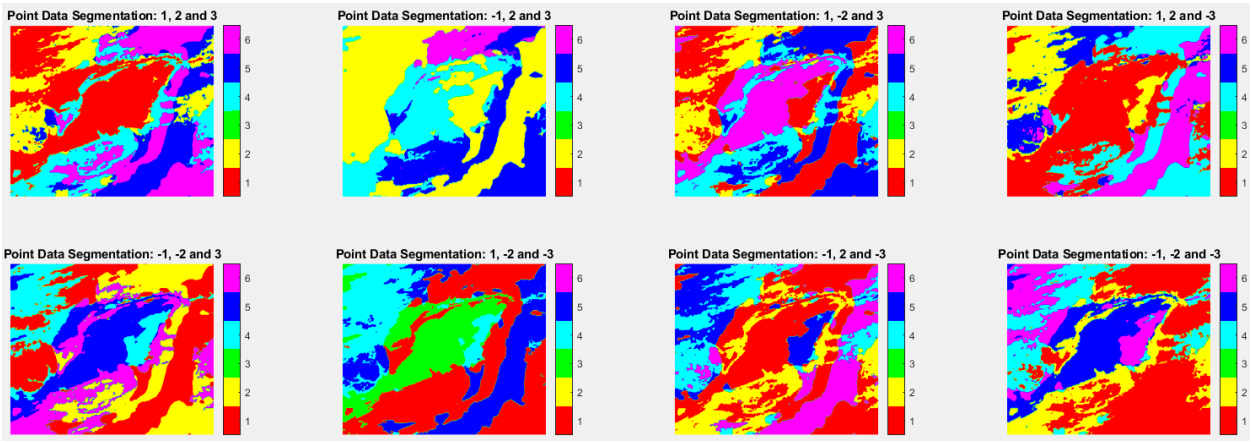
Segmented image before source separation.



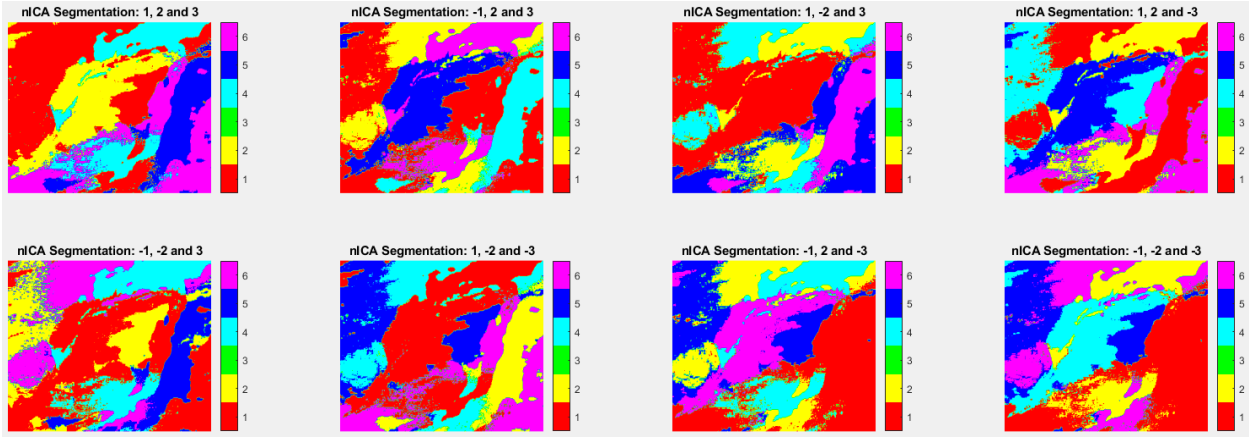
Segmented after nICA

Figure 52. An example of satellite image segmentation with SFE2D.

- An example of the segmentation on the point data sets (DEM, Gravity, and Magnetic data) is also presented below. As can be seen, spatial feature extraction has highlighted obscured structures in the segmented image.



Segmented point data sets (DEM, Gravity and Magnetic)



Segmentation after nICA

Figure 53. An example of point data sets segmentation with SFE2D.

11. Acknowledgments

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12. References

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Hyvärinen, **A**., **and E**. **Oja**, 2000, *Independent component analysis: algorithms and applications*, Neural Networks 13, 411–430.